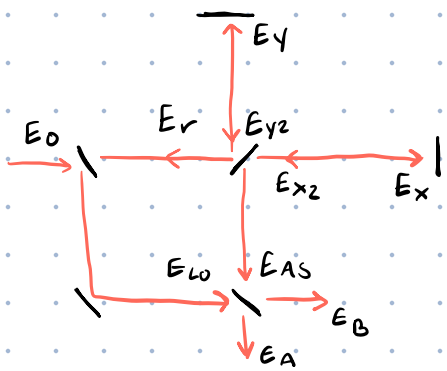
(i)	E_0	E_x	E_{x2}	E_y	E_{y2}	E_{LO}	E_{AS}	E_A	E_B	E_r
E_0	0	$t_{BS}e^{-i\phi_x}$	0	$-r_{BS}e^{-i\phi_y}$	0	$-r_{LO}e^{-i\phi_{HD}}$	0	0	0	0
E_x	0	0	$-r_x e^{-i\phi_x}$	0	0	0	0	0	0	0
E_{x2}	0	0	0	0	0	0	r_{BS}	0	0	t_{BS}
E_y	0	0	0	0	$-r_y e^{-i\phi_y}$	0	0	0	0	0
E_{y2}	0	0	0	0	0	0	t_{BS}	0	0	$-r_{BS}$
E_{LO}	0	0	0	0	0	0	0	r_{HD}	t_{HD}	0
E_{AS}	0	0	0	0	0	0	0	t_{HD}	$-r_{HD}$	0
E_A	0	0	0	0	0	0	0	0	0	0
E_B	0	0	0	0	0	0	0	0	0	0
E_r	0	0	0	0	0	0	0	0	0	0

* E_{LO} is defined as $-r_{LO}e^{-i\phi_{HD}}E_{in}$. E_{in} is $\frac{1}{2}E_0$ assuming the first beamsplitter is 50:50



1.2

$$E_{PDA} = r_{HD}E_{LO} + t_{HD}E_{AS}$$

$$= -r_{HD}r_{LO}e^{-i\phi_{HD}}E_{in} + t_{HD}(r_{BS}E_{x2} + t_{BS}E_{y2})$$

$$= -r_{HD}r_{LO}e^{-i\phi_{HD}}E_0 + t_{HD}(-r_{BS}r_x e^{-i\phi_x}E_x - t_{BS}r_y e^{-i\phi_y}E_y)$$

$$= -r_{HD}r_{LO}e^{-i\phi_{HD}}E_0 + t_{HD}(-r_{BS}t_{BS}r_x e^{-2i\phi_x} + t_{BS}r_{BS}r_y e^{-2i\phi_y})E_0$$

$$\boxed{\frac{E_{PDA}}{E_0} = -r_{HD}r_{LO}e^{-i\phi_{HD}} - t_{HD}r_{BS}t_{BS}r_x e^{-2i\phi_x} + t_{HD}t_{BS}r_{BS}r_y e^{-2i\phi_y}}$$

$$E_{PDB} = -t_{HD}r_{LO}e^{-i\phi_{HD}}E_0 - r_{HD}(-r_{BS}r_x t_{BS}e^{-2i\phi_x} + t_{BS}r_y r_{BS}e^{-2i\phi_y})E_0$$

$$\boxed{\frac{E_{PDB}}{E_0} = -t_{HD}r_{LO}e^{-i\phi_{HD}} + r_{HD}r_{BS}r_x t_{BS}e^{-2i\phi_x} - r_{HD}t_{BS}r_{BS}r_y e^{-2i\phi_y}}$$

1.3

$$\frac{E_A}{E_0} = -r_{HD}r_{LO}e^{-i\phi_{HD}} - \frac{1}{2}t_{HD}r_x e^{-2i\phi_x} + \frac{1}{2}t_{HD}r_y e^{-2i\phi_y}$$

$\phi_x = \phi_a + \phi_c$

$\phi_y = \phi_c - \phi_d$

$$\frac{E_A}{E_0} = -r_{HD} r_{LO} e^{-i\phi_{HD}} + \frac{1}{2} t_{HD}\left(-r_x e^{-2i(\phi_c+\phi_a)} + r_y e^{-2i(\phi_c-\phi_a)}\right)$$

*assume $r_x = r_y = 1$

$$= -r_{HD} r_{LO} e^{-i\phi_{HD}} + \frac{1}{2} t_{HD}\left(-\cancel{e^{-2i\phi_c}} - e^{-2i\phi_a} + \cancel{e^{-2i\phi_c}} + e^{2i\phi_a}\right)$$

$$= -r_{HD} r_{LO} e^{-i\phi_{HD}} - \frac{i}{2} t_{HD} e^{-i\phi_c} \sin(\phi_a)$$ → both have differential signs?

$$\frac{E_B}{E_0} = -t_{HD} r_{LO} e^{-i\phi_{HD}} - \frac{i}{2} r_{HD} e^{-i\phi_c} \sin(\phi_d)$$

1.4

$$P_A = \left(-r_{HD} r_{LO} e^{-i\phi_{HD}} - \frac{i}{2} t_{HD} e^{-i\phi_c} \sin(\phi_a)\right)\left(-r_{HD} r_{LO} e^{i\phi_{HD}} + \frac{i}{2} t_{HD} e^{i\phi_c} \sin(\phi_a)\right)$$

$$= r_{HD}^2 r_{LO}^2 (1) \underbrace{- \frac{i}{2} r_{HD} t_{HD} r_{LO} e^{i(\phi_c - \phi_{HD})} \sin(\phi_a) + \frac{i}{2} r_{HD} t_{HD} r_{LO} e^{i(\phi_{HD} - \phi_c)} \sin(\phi_a)}_{\frac{i}{2} r_{HD} t_{HD} r_{LO}\left(-e^{i(\phi_c - \phi_{HD})} + e^{i(\phi_{HD} - \phi_c)}\right) = \frac{i}{2} r_{HD} t_{HD} r_{LO} (2i \sin \phi_{HD} - \phi_c)}$$

$$+ \frac{1}{2} t_{HD}^2 \sin^2(\phi_a)$$

$$= r_{HD}^2 r_{LO}^2 + \frac{1}{2} t_{HD}^2 \sin^2(\phi_a) - r_{HD} t_{HD} r_{LO} \sin(\phi_{HD} - \phi_c) \sin(\phi_a)$$

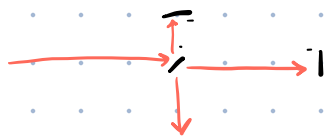
$$P_B = \left|\left(-t_{HD} r_{LO} e^{-i\phi_{HD}} - \frac{i}{2} r_{HD} e^{-i\phi_c} \sin(\phi_a)\right)\right|^2$$

$$= t_{HD}^2 r_{LO}^2 + \frac{1}{2} r_{HD}^2 \sin^2(\phi_a) - t_{HD} r_{HD} r_{LO} \sin(\phi_{HD} - \phi_c) \sin(\phi_a)$$

1.5

my answers above are certainly wrong but i don't know why/how
from above the signal depends on the darm phase and the difference btwn carm and HD.
if we pick a carm phase to maximize $\sin(\phi_{HD} - \phi_c)$, our signal is sensitive to ϕ_d
bc PDB-PDA leaves you with some term with $2\sin(\phi_d)$. this requires a
stable ϕ_{HD} & ϕ_c which is hard

2.1



	E_{in}	E_{x_1}	E_{x_2}	E_{y_1}	E_{y_2}	E_r	E_t
E_{in}	0	$t_{BS}e^{-i\phi_x}$	0	$-r_{BS}e^{-i\phi_y}$	0	0	0
E_{x_1}	0	0	$-r_x e^{-i\phi_x}$	0	0	0	0
E_{x_2}	0	0	0	0	0	t_{BS}	r_{BS}
E_{y_1}	0	0	0	0	$-r_y e^{-i\phi_y}$	0	0
E_{y_2}	0	0	0	0	0	$-r_{BS}$	t_{BS}
E_r	0	0	0	0	0	0	0
E_t	0	0	0	0	0	0	0

$$E_x = -r_x e^{-i\phi_x} E_{x_1}$$

$$\frac{E_x}{E_{in}} = -r_x t_{BS} e^{-2i\phi_x}$$

$$\frac{E_y}{E_{in}} = r_y r_{BS} e^{-2i\phi_y}$$

$$\frac{E_{as}}{E_{in}} = -r_{BS} r_x t_{BS} e^{-2i\phi_x} + t_{BS} r_y r_{BS} e^{-2i\phi_y}$$

2.2

$$\phi_x = \phi_x + \phi_x(t)$$
$$\phi_y = \phi_y + \phi_y(t)$$

$$\frac{E_x}{E_{in}} = -r_x t_{BS} \underbrace{e^{-2i(\phi_x + \Gamma\cos(\omega t))}}_{e^{-2i\phi_x}\left(\underbrace{e^{-2i\Gamma\cos(\omega t)}}_{1 - 2i\Gamma\underbrace{\cos(\omega t)}_{\frac{e^{i\omega t}+e^{-i\omega t}}{2}}}\right)}$$

$$\frac{E_x}{E_{in}} = -r_x t_{BS} e^{-2i\phi_x}\left(1 - i\Gamma\left(e^{i\omega t} + e^{-i\omega t}\right)\right)$$

* * *

these are the 3 fields

$$\frac{E_y}{E_{in}} = r_y r_{BS}\, e^{-2i\phi_y}\left(1 - i\Gamma\left(e^{i\omega t} + e^{-i\omega t}\right)\right)$$

2.3

$$\frac{E_{as}}{E_{in}} = r_{BS}\frac{E_x}{E_{in}} + t_{BS}\frac{E_y}{E_{in}}$$

$e^{-2i\phi_x} \to e^{-ik_0 L_x}$ where k_0 is $\frac{\omega_0}{c}$ ie carrier

$e^{-2i(\phi_y + \omega t)} \to e^{-2ik_+ L}$ where $k_+ = \frac{\omega_0 + \omega_+}{c}$ ← upper sideband

they do not accure the same phase bc length differences still matter

2.4

$$\frac{E_x}{E_{in}} = -\frac{1}{\sqrt{2}} r_x e^{-2i\phi_x}\left(1 - 2i\Gamma\cos\omega t\right)$$

$$\frac{E_y}{E_{in}} = \frac{1}{\sqrt{2}} r_y e^{-2i\phi_y}\left(1 - 2i\Gamma\cos(\omega t)\right)$$

$$\frac{E_{as}}{E_{in}} = -\frac{1}{2} r\, e^{-2i\phi_x}\left(1 - 2i\Gamma\cos\omega t\right) + \frac{1}{2} r e^{-2i\phi_y}\left(1 - 2i\Gamma\cos\omega t\right)$$

$$= -\frac{1}{2} r\left(e^{-2ik_0 L_x} - i\Gamma e^{-2ik_+ L_x + i\omega t} - i\Gamma e^{-2ik_- L_x - i\omega t}\right)$$

$$+ \frac{1}{2} r\left(e^{-2ik_0 L_y} - i\Gamma e^{-2ik_+ L_y + i\omega t} - i\Gamma e^{-2ik_- L_y - i\omega t}\right)$$

$$e^{-2i\phi_y} - e^{-2i\phi_x} = e^{-2i(\phi_c - \phi_a)} - e^{-2i(\phi_c + \phi_a)} = e^{-2i\phi_c} e^{2i\phi_a} - e^{-2i\phi_c} e^{-2i\phi_a}$$

$$= e^{-2i\phi_c}\left(e^{2i\phi_a} - e^{-2i\phi_a}\right)$$

$$= 2i e^{-2i\phi_c}\sin(2\phi_a)$$

$$e^{-2i\phi_x(t)} - e^{-2i\phi_y(t)} = -2i e^{-2i\phi_c}\, \underline{e^{-2i\omega L_c/c}}\, \sin\left(2\phi_a + 2\underline{\frac{\omega L_a}{c}}\right)$$

its the same as above but with the extra sideband

$$\frac{E_{AS}}{E_{in}} = \frac{1}{\cancel{2}}\Big[-r\Big(\cancel{2}i e^{-2i\phi_c}\sin(2\phi_a) - \cancel{2}i e^{-2i\phi_c} e^{-2i\omega L_c/c}\sin\left(2\phi_a + 2\frac{\omega L_a}{c}\right)$$

$$\underbrace{- \cancel{2}i e^{-2i\phi_c} e^{2i\omega L_c/c}\sin\left(2\phi_a - 2\frac{\omega L_a}{c}\right)}$$

$$+ r\left(\text{same expression for } L_y\right)\Big]$$

$$= r_x\left[\sin(2\phi_a) - \Gamma\sin(2\phi_a)\sin\left(\frac{\omega\phi_a}{\omega_0}\right)\sin\left(\omega t + \frac{\omega\phi_c}{\omega_0}\right)\right]$$

$$\frac{P_{as}}{P_{in}} = r_x^2\left[\sin^2(2\phi_a) - 2\sin(4\phi_a)\sin\left(\frac{\omega\phi_a}{\omega_0}\right)\sin\left(\omega t + \frac{\omega\phi_c}{\omega_0}\right)\right]$$

Γ^2 term disappears

2.5

over one cycle $\sin(\omega t) \to 0$

$$P_{AS}^{DC} = r_x^2 \sin^2(2\phi_a)$$

ie standard mich as power

over one cycle $\sin\left(\omega t + \frac{\omega\phi_a}{\omega_0}\right)\cos(\omega t) \to \frac{1}{2}\sin\left(\frac{\omega\phi_a}{\omega_0}\right)$

$$P_{AS}^{I} = -r_x^2 \sin\left(\frac{\omega\phi_c}{\omega_0}\right)\Gamma\sin(4\phi_a)\sin\left(\frac{\omega\phi_a}{\omega_0}\right)$$

* added i in plot

over one cycle $\sin\left(\omega t + \frac{\omega\phi_a}{\omega_0}\right)\sin(\omega t) \to \frac{1}{2}\cos\left(\frac{\omega\phi_a}{\omega_0}\right)$

$$P_{AS}^{Q} = -r_x^2 \cos\left(\frac{\omega\phi_c}{\omega_0}\right)\Gamma\sin(4\phi_a)\sin\left(\frac{\omega\phi_a}{\omega_0}\right)$$

2.6

on github